Lecture 4 v2:
OLS Implementation

25117 - Econometrics

Universitat Pompeu Fabra

October 9th, 2024

What we learned in the last lesson

- The population regression line, $\beta_0 + \beta_1 X$, is the mean of Y as a function of the value of X.
  - The slope, β_1, is the expected difference in Y between two observations with X values that differ by one unit.
  - The intercept, β_0, determines the level (or height) of the regression line.

- The population regression line can be estimated using sample observations (Y_i, X_i) with $i = 1, \ldots, n$ by ordinary least squares (OLS).
  - The OLS estimators of the regression intercept and slope are denoted $\hat{\beta}_0$ and $\hat{\beta}_1$.
  - The predicted value of Y given X is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$.

What we learned in the last lesson

- The R^2 and standard error of the regression (SER) are measures of how close the values of Y_i are to the estimated regression line.
  - The R^2 is between 0 and 1, indicating how much variance in our outcome is explained by the variance of our regressors.
  - The standard error of the regression estimates the standard deviation of the regression error.

- There are three key assumptions for estimating causal effects using the linear regression model:
  1. The regression errors, u_i, have a mean of 0, conditional on the regressors X_i;
  2. the sample observations are i.i.d. random draws from the population; and
  3. large outliers are unlikely

- If these assumptions hold, the OLS estimator $\hat{\beta}_1$ is
  1. an unbiased estimator of the causal effect β_1
  2. consistent
  3. normally distributed when the sample is large.

Estimation of the regression line

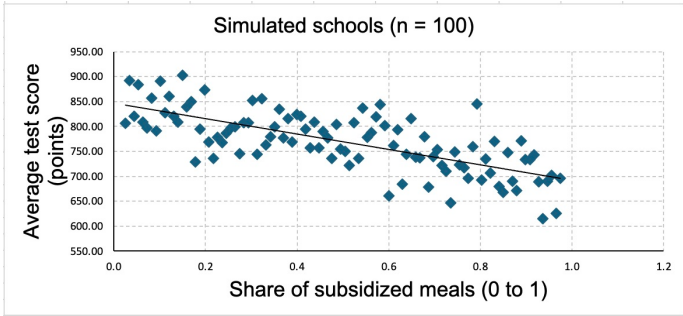
We want to learn about the slope of the population regression line. We have data from a sample, so there is sampling uncertainty. There are five steps toward this goal:

1. State the population of interest
2. Provide an estimator of this population object
3. Derive the sampling distribution of the estimator (this requires certain assumptions). In large samples, this sampling distribution will be normal by the CLT.
4. The square root of the estimated variance of the sampling distribution is the standard error (SE) of the estimator
5. Use the SE to construct t-statistics (for hypothesis tests) and confidence intervals.

Estimation of the regression line

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

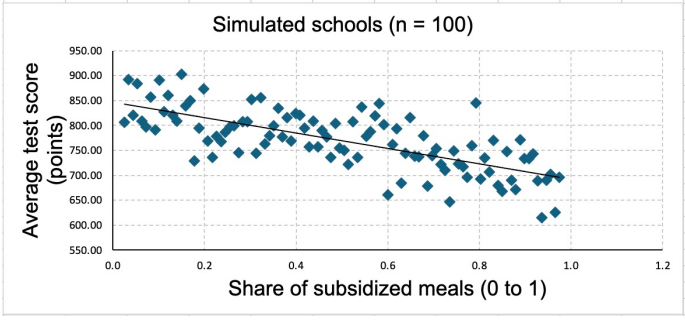
- β_1 is the slope of the population regression line
- $\hat{\beta}_1$ is the OLS estimator of β_1
- Under the maintained OLS assumptions, the large-sample distribution conditional on X is approximately $\mathcal{N}(\beta_1; \sigma_u^2 / TSS_X)$.



Here $TSS_X = \sum_{i=1}^{n} (X_i - \overline{X})^2$ is the **total sum of squared deviations of X from its sample mean**.

Hypothesis Testing

- Most of the time, we test for $H_0 : \beta_1 = 0$
- Null hypothesis and two-sided alternative:
  - $H_0 : \beta_1 = \beta_{1,0}$ vs. $H_1 : \beta_1 \neq \beta_{1,0}$
- Null hypothesis and one-sided alternative:
  - $H_0 : \beta_1 = \beta_{1,0}$ vs. $H_1 : \beta_1 < (or >)\beta_{1,0}$



Hypothesis Testing

- In general, the t-statistic is given by

$$t = \frac{\text{estimator} - \text{hypothetised value}}{\text{standard error of the estimator}}$$

- For μ_Y, the t-statistic is given by

$$t = \frac{\overline{Y} - \mu_{Y,0}}{s_Y/\sqrt{n}}$$

- For β_1, the t-statistic is given by

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)}$$

where $SE(\hat{\beta}_1) = \sqrt{\frac{SSR}{(n-2)TSS_X}} = \sqrt{\frac{\sum_{i=1}^n \hat{u}_i^2}{(n-2)\sum_{i=1}^n (X_i - \overline{X})^2}}$.

- Using the large-sample normal approximation, reject $H_0 : \beta_1 = \beta_{1,0}$ at 5% if $|t| > 1.96$, or if the two-sided p-value is below 0.05. Excel's finite-sample test uses a t_{n-2} critical value instead.

Hypothesis Testing: The Intercept

- To test the intercept, specify

$$H_0 : \beta_0 = \beta_{0,0}, \text{ against } H_1 : \beta_0 \neq \beta_{0,0}.$$

- The t-statistic is

$$t = \frac{\hat{\beta}_0 - \beta_{0,0}}{SE(\hat{\beta}_0)},$$

where

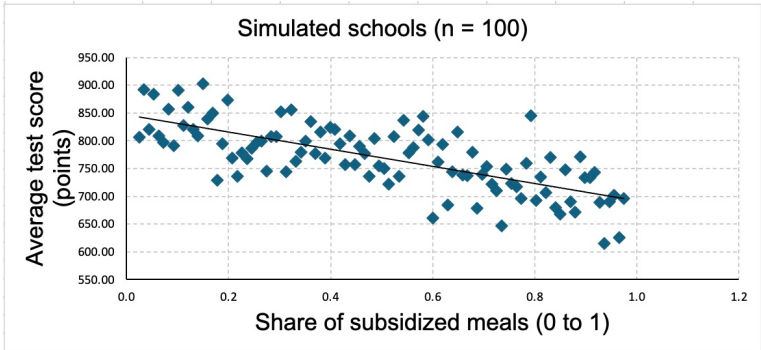
$$SE(\hat{\beta}_0) = SER\sqrt{\frac{1}{n} + \frac{\overline{X}^2}{TSS_X}} = \sqrt{\frac{SSR}{n-2}\left(\frac{1}{n} + \frac{\overline{X}^2}{TSS_X}\right)}.$$

- For $H_0 : \beta_0 = 0$, this simplifies to $t = \hat{\beta}_0 / SE(\hat{\beta}_0)$: we test whether the population regression line's value at $X = 0$ is zero.
- Using the large-sample normal approximation, reject H_0 at 5% if $|t| > 1.96$, or if the two-sided p-value $2[1 - \Phi(|t|)]$ is below 0.05. Otherwise, do not reject H_0.
- Excel's finite-sample test uses a t_{n-2} critical value instead.

Excel: A Dataset of 100 Simulated Schools

Open `Lecture4_Excel_100schools.xlsx`, sheet **Data**.

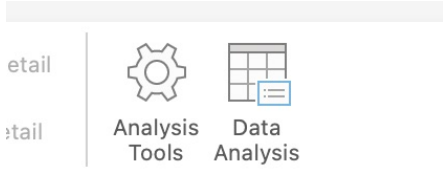
	A	B	C
1	Meal_share	Test_score	School_ID
2	0.0826	856.56	1
3	0.7447	749.42	2
4	0.1497	903.17	3
5	0.8311	770.40	4
6	0.8023	692.62	5
7	0.4760	736.09	6
8	0.4088	819.93	7
9	0.5432	836.38	8
10	0.3801	815.97	9
11	0.1402	808.97	10
12	0.4472	756.63	11
13	0.7927	844.84	12
14	0.8407	679.51	13
15	0.5816	844.02	14
16	0.1306	819.90	15
17	0.7063	753.72	16
18	0.6871	678.74	17
19	0.7159	722.32	18
20	0.4568	790.13	19
21	0.6775	779.54	20
22	0.5144	722.22	21
23	0.4952	754.44	22



- X_i: share of subsidized meals (0–1). Y_i: average test score, in points.
- These are **simulated teaching data**, with a negative association similar to the earlier example. They are not the original school records.

Excel Step 1: Enable the Analysis ToolPak

1. In desktop Excel, enable **Analysis ToolPak** if **Data Analysis** is missing:
   - **Mac:** Tools → Excel Add-ins → Analysis ToolPak → OK.
   - **Windows:** File → Options → Add-ins; Manage: Excel Add-ins → Go. Check Analysis ToolPak and click OK.
2. Open the **Data** tab and click **Data Analysis**.



The screenshots show Excel for Mac. The regression input ranges and calculations are the same on Windows.

Setup reference: Microsoft: Load the Analysis ToolPak in Excel.

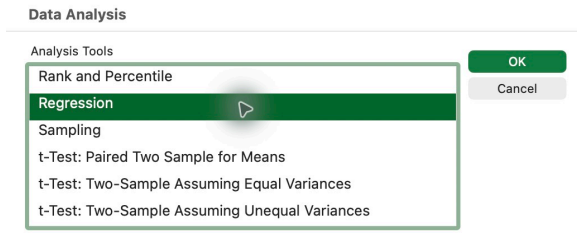
Excel Step 2: Choose Regression

1. In the list of analysis tools, select **Regression**.
2. Click **OK**.

We want Excel to estimate

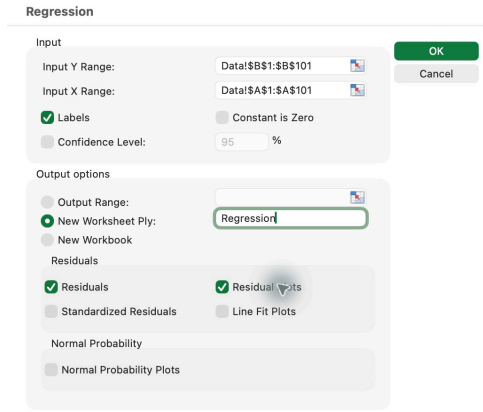
$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

by minimizing the sum of squared residuals.



Excel Step 3: Set the Input Ranges

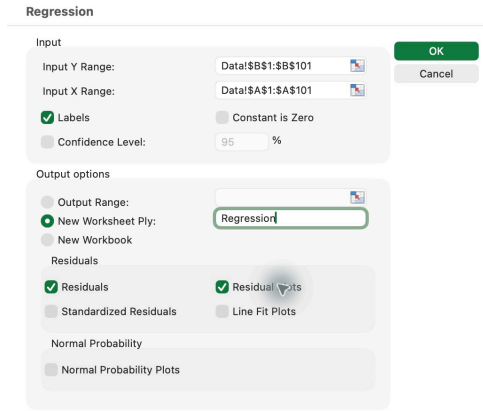
1. **Input Y Range:**
   `Data!$B$1:$B$101`
   The outcome is **Test_score**.
2. **Input X Range:**
   `Data!$A$1:$A$101`
   The explanatory variable is **Meal_share**.
3. Check **Labels**: row 1 contains names, so there are 100 observations.
4. Leave **Constant is Zero** unchecked: estimate an intercept.



Excel Step 4: Choose the Output

1. Choose **New Worksheet Ply** and name it **Regression**. If that sheet already exists, use a new name, such as **Regression2**.
2. Check **Residuals** and **Residual Plots**.
3. The output includes **95% confidence intervals** by default.
4. Click **OK**. Widen the output columns if numbers are cut off.

ToolPak output is a **snapshot**: rerun the regression after changing the data.



Excel Step 5: Read the Summary Output

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.713178796
R Square	0.508623996
Adjusted R Square	0.503609955
Standard Error	42.59996002
Observations	100

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	184088.797	184088.797	101.4399384	8.45674E-17
Residual	98	177846.1462	1814.756594		
Total	99	361934.9432			

- **Observations:** $n = 100$. Two coefficients are estimated, so residual degrees of freedom are $100 - 2 = 98$.
- **R Square:** 0.5086. **Standard Error:** 42.60, the SER.
- Under ANOVA, **Residual SS** is our *SSR*; **Total SS** is TSS_Y.

Yellow cells identify the results discussed in these slides.

Excel: Interpret the Estimated Coefficients

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	847.0719189	8.790754329	96.35941209	6.7995E-99	829.6269531	864.5168848
Meal_share	-154.8950979	15.37918018	-10.07173959	8.45674E-17	-185.4145804	-124.3756153

$$\widehat{Y}_i = 847.0719 - 154.8951 X_i.$$

- **Intercept:** the fitted score at $X = 0$ is 847.07. Since the simulated sample has $X > 0$, this is an extrapolation.
- **Slope:** increasing X by 0.10 (10 percentage points) is associated with a change of

$$-154.8951 \times 0.10 = -15.49 \text{ test-score points.}$$

- An association alone does not establish a causal effect of subsidized meals.

Excel: Test Whether Each Coefficient Is Zero

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	847.0719189	8.790754329	96.35941209	6.7995E-99	829.6269531	864.5168848
Meal_share	-154.8950979	15.37918018	-10.07173959	8.45674E-17	-185.4145804	-124.3756153

$$H_0 : \beta_j = 0, \qquad t_j = \frac{\widehat{\beta}_j - 0}{SE(\widehat{\beta}_j)}, \qquad p_j = P(|T_{98}| \geq |t_j|)\,.$$

Coefficient	Standard error	t statistic	Two-sided p-value
Intercept	8.7908	96.3594	6.80×10^{-99}
Meal_share	15.3792	-10.0717	8.46×10^{-17}

Both p-values are below 0.05: reject both zero-coefficient null hypotheses at 5%. These are very small probabilities, **not zero**.

Exact finite-sample t_{98} inference additionally assumes conditionally normal errors, with independent observations and conditional mean zero.

Excel: Construct 95% Confidence Intervals

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	847.0719189	8.790754329	96.35941209	6.7995E-99	829.6269531	864.5168848
Meal_share	-154.8950979	15.37918018	-10.07173959	8.45674E-17	-185.4145804	-124.3756153

$$\widehat{\beta}_j \;\pm\; t_{0.975,98}\, SE(\widehat{\beta}_j), \qquad t_{0.975,98} = 1.9845.$$

$$CI_{\beta_0} = 847.0719 \pm 1.9845(8.7908) = [829.6270,\ 864.5169],$$
$$CI_{\beta_1} = -154.8951 \pm 1.9845(15.3792) = [-185.4146,\ -124.3756].$$

- These are Excel's **Lower 95%** and **Upper 95%** columns (using unrounded values).
- Neither interval contains zero, consistent with the two-sided tests.
- In repeated sampling, 95% of intervals constructed this way cover the true coefficient, under the stated assumptions.

Excel: SER and RMSE

First compute the fitted value and the residual:

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i, \qquad \widehat{u}_i = Y_i - \widehat{Y}_i, \qquad SSR = \sum_{i=1}^{100} \widehat{u}_i^2 = 177{,}846.1462.$$

Excel's Standard Error = SER

$$SER = \sqrt{\frac{SSR}{n-2}} = \sqrt{\frac{177{,}846.1462}{98}} = 42.6000.$$

The denominator adjusts for estimating two coefficients. This estimates the error standard deviation.

RMSE with denominator n

$$RMSE = \sqrt{\frac{SSR}{n}} = \sqrt{\frac{177{,}846.1462}{100}} = 42.1718.$$

This is the square root of the average squared prediction error within the sample.

Both are measured in **test-score points**. In this course, distinguish the denominators; the earlier Stata Root MSE label corresponds to the SER.

Excel: Compute and Interpret R^2

$$TSS_Y = \sum_i (Y_i - \overline{Y})^2 = 361{,}934.9432,$$
$$SSR = \sum_i (Y_i - \widehat{Y}_i)^2 = 177{,}846.1462,$$
$$ESS = \sum_i (\widehat{Y}_i - \overline{Y})^2 = 184{,}088.7970.$$

With an intercept, $TSS_Y = ESS + SSR$. Therefore,

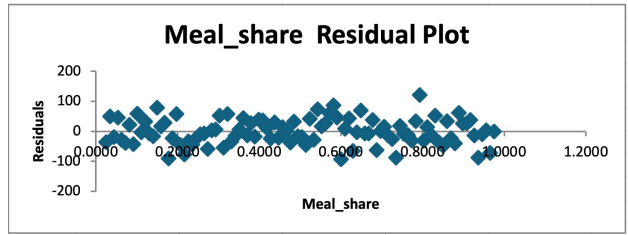
$$R^2 = 1 - \frac{SSR}{TSS_Y} = 1 - \frac{177{,}846.1462}{361{,}934.9432} = 0.5086.$$

- The fitted line explains **50.86% of the sample variation** in test scores around their mean.
- This measures in-sample fit, not the proportion of scores caused by meals.

Excel: Inspect Predicted Values and Residuals

RESIDUAL OUTPUT

Observation	Predicted Test_score	Residuals
1	834.2813013	22.27669867
2	731.7220042	17.69599576
3	823.8768427	79.29015729
4	718.344644	52.05135599
5	722.8037641	-30.18276409
6	773.3403034	-37.2543034
7	783.7449169	36.18008309
8	762.9356899	73.44031012
9	788.2038821	27.76511791
10	825.3630612	-16.38906117



For observation 1, $Y_1 = 856.5580$ and $\widehat{Y}_1 = 834.2813$, so

$$\widehat{u}_1 = Y_1 - \widehat{Y}_1 = 22.2767.$$

The model underpredicts this school's score by 22.28 points. Look for systematic patterns in the residuals.

By Hand, Step 1: Means and Deviations

Open **Paso_a_paso**. Each of the 100 observations has its own row (43–142).

$$\overline{X} = \frac{\sum_i X_i}{100} = 0.5000, \qquad \overline{Y} = \frac{\sum_i Y_i}{100} = 769.6244.$$

i	Xi	Yi	Xi - X-bar	Yi - Y-bar	(Xi-Xbar)^2	(Yi-Ybar)^2	(Xi-Xbar)(Yi-Ybar)
1	0.0826	856.5580	-0.4174	86.9336	0.1742	7,557.4560	-36.2882
2	0.7447	749.4180	0.2447	-20.2064	0.0599	408.2974	-4.9444
3	0.1497	903.1670	-0.3503	133.5426	0.1227	17,833.6340	-46.7737
4	0.8311	770.3960	0.3311	0.7716	0.1096	0.5954	0.2555
5	0.8023	692.6210	0.3023	-77.0034	0.0914	5,929.5190	-23.2760
6	0.4760	736.0860	-0.0240	-33.5384	0.0006	1,124.8223	0.8046
7	0.4088	819.9250	-0.0912	50.3006	0.0083	2,530.1534	-4.5855
8	0.5432	836.3760	0.0432	66.7516	0.0019	4,455.7801	2.8825

- Row 43: `D43=B43-$C$5` computes $X_1 - \overline{X}$; `E43=C43-$C$6` computes $Y_1 - \overline{Y}$.
- Then square each deviation and multiply the two deviations: `F43=D43^2`, `G43=E43^2`, `H43=D43*E43`.
- Copy down through row 142. The dollar signs keep the means fixed. Row 144 sums each column; both deviation sums are zero up to rounding.

By Hand, Step 2: From Sums to the OLS Line

Step	Concept	Value	Excel formula / explanation
1	Sample size n	100	=COUNT(Data!A2:A101)
2	Mean X (X-bar)	0.5000	=AVERAGE(Data!A2:A101)
3	Mean Y (Y-bar)	769.6244	=AVERAGE(Data!B2:B101)
4	Sum (Xi - X-bar)	0.0000	=SUM(D43:D142)
5	Sum (Yi - Y-bar)	0.0000	=SUM(E43:E142)
6	TSS_X	7.6728	=SUM(F43:F142)
7	Cross-product sum	-1,188.4740	=SUM(H43:H142)
8	Sample variance s_X^2	0.0775	=C9/(C4-1)
9	Sample covariance s_XY	-12.0048	=C10/(C4-1)
10	Slope b1-hat	-154.8951	=C10/C9
11	Intercept b0-hat	847.0719	=C6-C13*C5

Let $S_{XY} = \sum_i (X_i - \overline{X})(Y_i - \overline{Y})$ be the **cross-product sum**. Also, $TSS_X = \sum_i (X_i - \overline{X})^2$ is the **total sum of squared deviations of X from its sample mean**. Then $s_{XY} = S_{XY}/(n-1)$ and $s_X^2 = TSS_X/(n-1)$.

$$\widehat{\beta}_1 = \frac{S_{XY}}{TSS_X} = \frac{s_{XY}}{s_X^2} = \frac{-1188.4740}{7.6728} = -154.8951,$$

$$\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X} = 769.6244 - (-154.8951)(0.5000) = 847.0719.$$

Use `C13=C10/C9` and `C14=C6-C13*C5`. Keep full precision in the cells.

By Hand, Step 3: Predictions, Fit, and Error Size

For the first observation (row 43), enter:

Cell	Excel formula	Meaning
I43	`=$C$14+$C$13*B43`	$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i$
J43	`=C43-I43`	$\widehat{u}_i = Y_i - \widehat{Y}_i$
K43	`=J43^2`	$\widehat{u}_i^2$
L43	`=(I43-$C$6)^2`	$(\widehat{Y}_i - \overline{Y})^2$

Copy down through row 142. Sum column K for *SSR*, G for TSS_Y, and L for *ESS*.

12	SSR (sum residual^2)	177,846.1462	=SUM(K43:K142)
13	TSS_Y	361,934.9432	=SUM(G43:G142)
14	ESS (explained SS)	184,088.7970	=SUM(L43:L142)
15	R Square	0.5086	=1-C15/C16
16	Residual degrees of freedom	98	=C4-2
17	SER (Excel Standard Error)	42.6000	=SQRT(C15/C19)
18	RMSE (denominator n)	42.1718	=SQRT(C15/C4)

Then calculate $R^2 = 1 - SSR/TSS_Y$, $SER = \sqrt{SSR/98}$, and $RMSE = \sqrt{SSR/100}$.

By Hand, Step 4: Standard Errors and t Statistics

Under the maintained OLS assumptions, use SER^2 to estimate σ_u^2:

$$SE(\widehat{\beta}_1) = \frac{SER}{\sqrt{TSS_X}} = \frac{42.6000}{\sqrt{7.6728}} = 15.3792,$$

$$SE(\widehat{\beta}_0) = SER\sqrt{\frac{1}{n} + \frac{\overline{X}^2}{TSS_X}} = 42.6000\sqrt{\frac{1}{100} + \frac{0.5000^2}{7.6728}} = 8.7908.$$

In **Paso_a_paso**, the formulas are:

- C22: `=C20/SQRT(C9)`; C23: `=C20*SQRT(1/C4+C5^2/C9)`.
- To test zero coefficients, divide each estimate by its own standard error:

$$t_1 = \frac{-154.8951}{15.3792} = -10.0717, \qquad t_0 = \frac{847.0719}{8.7908} = 96.3594.$$

- C24: `=C13/C22`; C25: `=C14/C23`.